

Sugam Bhattarai

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PROFESSIONAL SUMMARY

Engineering Physics senior specializing in mechanical applications with hands-on experience in mechanical design, CAD, instrumentation, automated testing, experimental research, data analysis, and technical documentation. Experienced with SOLIDWORKS, AutoCAD, LabVIEW, NI-DAQ, Python, and MATLAB, with strengths in troubleshooting, experimentation, documentation, and technical communication.

EDUCATION

Southeast Missouri State University

Cape Girardeau, Missouri

B.S. in Engineering Physics: Mechanical Application; Minor in Physics

Expected Dec. 2026

GPA: 3.892/4.000

TECHNICAL SKILLS

Mechanical Design & CAD: SOLIDWORKS, AutoCAD, 3D modeling, parts and assemblies, engineering drawings, component layouts, bill of materials, 3D printing

Testing & Instrumentation: LabVIEW, NI-DAQ, NI USB-6003, automated measurements, circuit testing, experimental setup, troubleshooting, validation, repeatability testing

Programming & Data Analysis: Python, MATLAB, C++, pandas, matplotlib, SciPy, Microsoft Excel, linear regression, experimental data analysis

Laboratory & Research Methods: HUVEC culture, sterile technique, media preparation, cell splitting, cell treatment, fixation, photobiomodulation, biomaterial studies, experimental documentation

ENGINEERING & TECHNICAL PROJECTS

Automated Electric Circuit Analysis Using DAQ and Python

Aug. 2025 – Dec. 2025

Undergraduate Laboratory Enhancement Project, Southeast Missouri State University

- Built a LabVIEW and NI USB-6003 system for automated voltage sweeps and computer-based data collection.
- Characterized current-voltage behavior of resistors and red, blue, green, and infrared LEDs.
- Generated 45+ datasets across 15 resistance values and multiple diode trials using repeatable testing procedures.
- Analyzed data with Python, pandas, matplotlib, and SciPy; resistor fits achieved $R^2 > 0.99$.

One-Gallon Stainless-Steel Enclosure Assembly

Mar. 2023 – Apr. 2023

- Designed a Type 316 stainless-steel enclosure with a removable panel and corrosion-resistant hardware.
- Produced 2D engineering drawings, orthographic and isometric views, and an exploded assembly.
- Created a bill of materials with item balloons, specifications, dimensions, and assembly documentation.
- Integrated 10-32 PEM nuts, a gasket, Schrader valve, pressure gauge, and fastening hardware.

RESEARCH & TECHNICAL EXPERIENCE

Undergraduate NSF ERI Researcher

Aug. 2025 – Present

Department of Physics and Chemistry, Southeast Missouri State University

Cape Girardeau, Missouri

- Study HUVEC recovery following experimentally induced cellular damage under controlled treatment conditions.
- Induce cellular stress with hydrogen peroxide and evaluate recovery using laminin, collagen, and red/blue light.
- Conduct ongoing work on a glucose-based injury model and gold nanoparticle biocompatibility for therapeutic applications.
- Perform sterile cell culture, media preparation, cell splitting, treatment, observation, fixation, and documentation.
- Analyze cellular response and experimental limitations while supporting development of a therapeutic-device concept.

Teaching Assistant

Aug. 2024 – May 2026

Department of Physics and Chemistry, Southeast Missouri State University

Cape Girardeau, Missouri

- Guided students through physics laboratories, measurements, calculations, and data analysis.
- Prepared and maintained laboratory equipment while troubleshooting setup, wiring, and measurement issues.
- Graded assignments and laboratory reports with feedback on calculations, technical writing, and experimental reasoning.

Learning Assistant

Sept. 2024 – Present

Learning Assistance Program, Southeast Missouri State University

Cape Girardeau, Missouri

- Tutor students in physics, engineering physics, mathematics, and quantitative problem-solving.
- Explain technical concepts through structured approaches adapted to individual student needs.

ADDITIONAL PROFESSIONAL EXPERIENCE

Resident Assistant

Office of Residence Life, Southeast Missouri State University

Apr. 2024 – Present
Cape Girardeau, Missouri

- Support residents with academic and personal concerns and connect them with appropriate campus resources.
- Respond to emergencies, conflicts, safety concerns, and facility issues while documenting incidents and coordinating follow-up.

Summer Operations Assistant

Office of Residence Life, Southeast Missouri State University

May 2026 – Aug. 2026
Cape Girardeau, Missouri

- Supported camp and conference housing through check-in, room setup, building preparation, and housing logistics.
- Responded to guest and facility needs while coordinating daily operations with Residence Life staff.

Orientation Leader

New Student Programs, Southeast Missouri State University

Sept. 2023 – Oct. 2024
Cape Girardeau, Missouri

- Led campus tours and introduced students and families to academic expectations, services, and university resources.
- Answered questions and helped incoming students navigate campus and transition to university life.

AWARDS & ACHIEVEMENTS

- **First Place, Graduate Poster Presentation**, 34th Annual Research Conference, Southeast Missouri State University, 2026
- **Administrative Excellence Award**, Office of Residence Life, Southeast Missouri State University, 2025–2026
- **President's List** and **Dean's List**, Southeast Missouri State University, multiple semesters

CERTIFICATIONS

- **SOLIDWORKS 2022 Essential Training**, LinkedIn Learning, Sept. 2026
- **AutoCAD: 3D Modeling for Mechanical Designs**, LinkedIn Learning, Sept. 2026

PROFESSIONAL MEMBERSHIP

- **Biomedical Engineering Society (BMES)**, Student Member, Sept. 2026 – Present